

Combinatorics PhD Exam, August 2005

Show your work.

1. This problem has two parts.

- a. There are r black and $n - r$ white balls in an urn. They are removed one at a time without replacement. What is the probability that exactly k drawings are required to get a white ball?
- b. Use your result to conclude that

$$\sum_{k=1}^r \frac{(r)_k}{(n-1)_k} = \frac{r}{n-r}.$$

2. Each of n people is to be mailed an envelope containing a letter and a bill. How many ways Q_n are there of placing the n letters and n bills into n addressed envelopes so that no envelope contains both the correct letter and bill?

3. Let P_n be the total number of k -permutations of n for various k , that is,

$$P_n = \sum_{k=0}^n (n)_k, \quad n = 0, 1, \dots$$

Show that

$$P(t) = \sum_{n=0}^{\infty} P_n \frac{t^n}{n!} = (1-t)^{-1} e^t$$

and use this to show that

$$P_n = nP_{n-1} + 1, \quad n = 1, 2, \dots, \quad P_0 = 1.$$

4. Define the binary Hamming code $H(r)$ of length $2^r - 1$. Show that $H(r)$ is an exactly single error correcting code and that $H(r)$ is a perfect code. Determine the number of codewords of weight 3 in $H(r)$.
5. Show that the number of partitions of a number n into exactly m parts is equal to the number of partitions of $n - m$ into no more than m parts.
6. An order on the set of ordered pairs of non-negative integers is defined by $(a_1, a_2) \leq (b_1, b_2)$ if $a_i \leq b_i$ for $i = 1, 2$. Find the Möbius function of this poset.
7. Determine for which values of m and n the complete bipartite graph K_{mn} is. Answer the same three questions for the n -cube. (Recall that the n -cube Q_n is defined as the graph whose vertices are the set of all binary sequences of length n , where two vertices are adjacent if the corresponding sequences differ by exactly one digit.)
 - (a) planar
 - (b) Eulerian
 - (c) Hamiltonian.
8. Let G be a triangle-free graph with n vertices, minimum degree k and girth g . Prove that $g \leq 2n/k$.
Hint: For an appropriate cycle C count the number of edges from C to $G \setminus C$ in two ways.